

Predicting the health of Lithium ion Batteries using Machine Learning.

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Objective & Research Question:

Can machine learning models, specifically Gaussian Process Regression (GPR), accurately predict lithium-ion battery health using Electrochemical Impedance Spectroscopy (EIS) data to improve efficiency and lifespan?

Background:

- Lithium-ion batteries are widely used in electronics, electric vehicles, and grid storage.
- Battery degradation over time affects performance and safety.
- Traditional health diagnostics like capacity fade tests are expensive and time-consuming.
- Machine learning offers a data-driven, scalable alternative for battery health monitoring.
- GPR is ideal for modeling complex, non-linear degradation patterns.

What is EIS?

Electrochemical Impedance Spectroscopy (EIS) is a diagnostic technique that applies small alternating current (AC) signals at varying frequencies to a battery and measures its response. It reveals electrochemical properties like resistance and capacitance — giving insight into battery health **without requiring full charge-discharge cycles**.

This project uses **real EIS data collected from commercial-grade lithium-ion batteries** undergoing realistic usage and degradation.

Methods:

- **Data Source:** EIS data from commercial batteries at different charge-discharge cycles.
- **Preprocessing:** Filtered frequencies (0.2–20,000 Hz), dropped null values and cycle 0
- **Feature Engineering:** Real and imaginary impedance ($\text{Re}(Z)$, $\text{Im}(Z)$) paired at each frequency as model features.
- **Model:** Gaussian Process Regressor (with RBF kernel) trained on standardized impedance features.
- **Comparison:** GPR model based on EIS vs. a cycle-based GPR baseline.

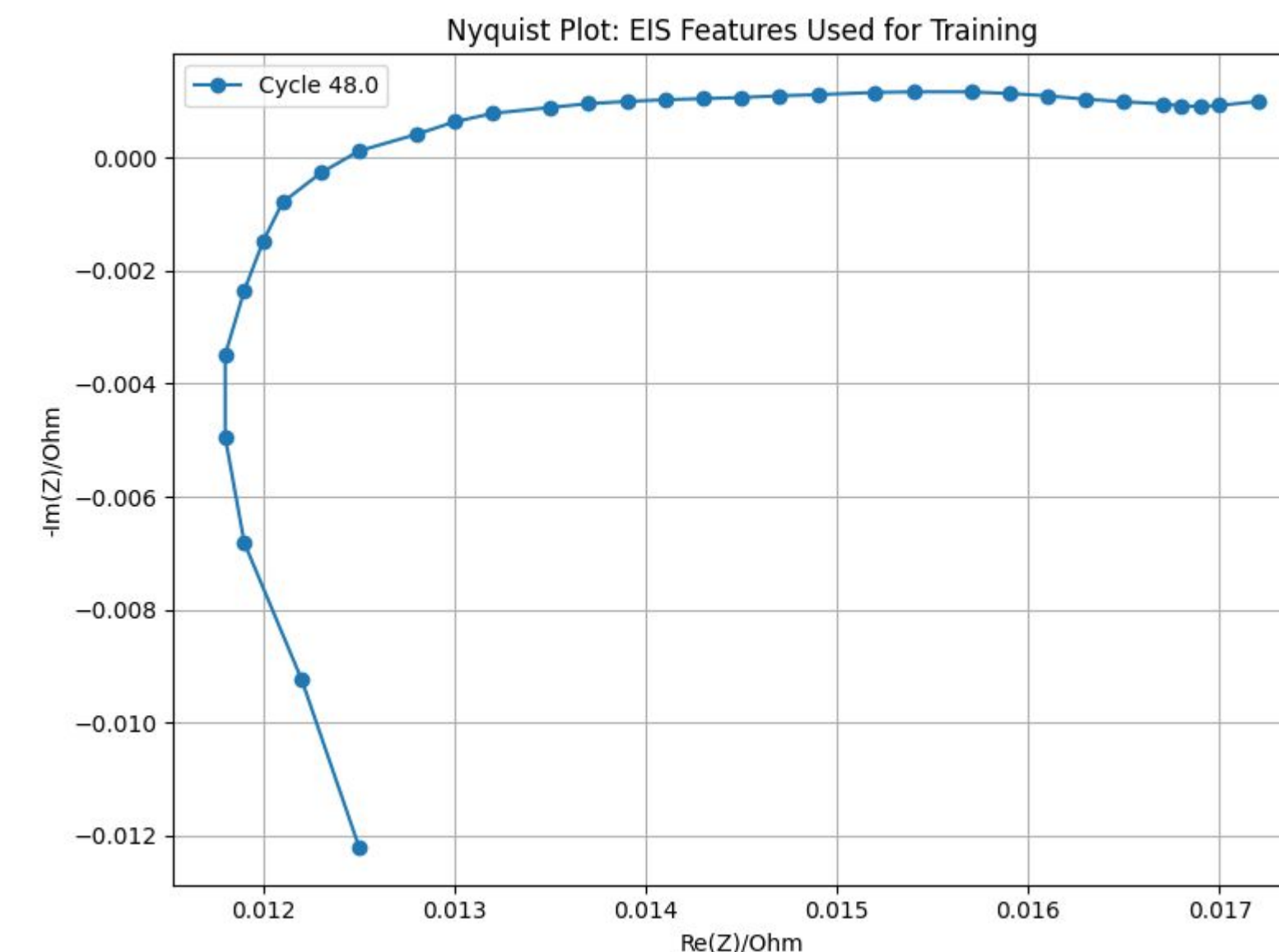


Fig 1: Nyquist Plot of Impedance Evolution Across Cycles

Description: Shows how impedance changes as the battery ages.

Explanation: Curve shape shifts with cycles, revealing degradation patterns used as model features.

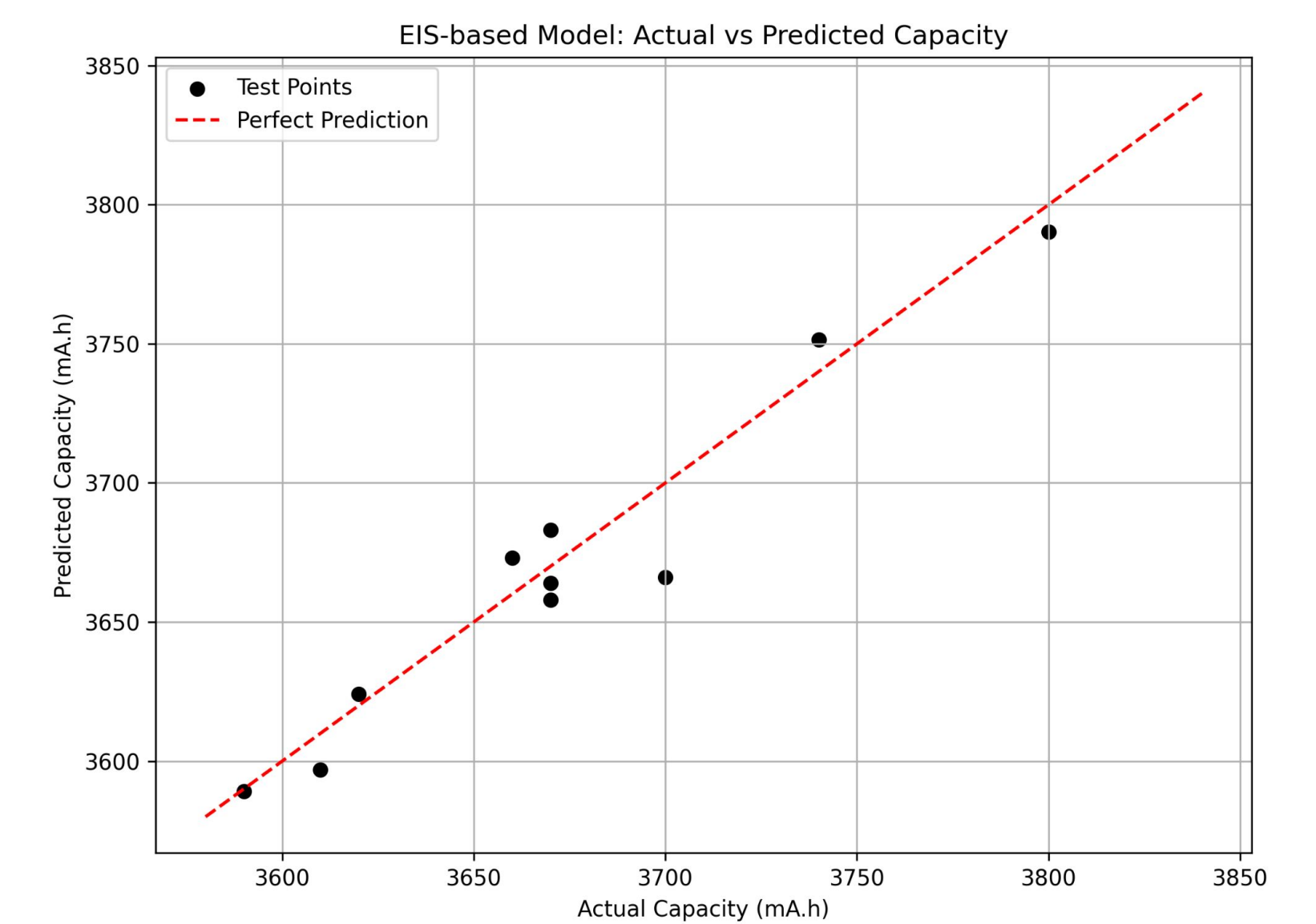


Fig 2: EIS-based GPR — Actual vs Predicted Capacity

Description: Compares predicted vs actual battery capacity.

Explanation: Strong alignment with the diagonal line shows high model accuracy

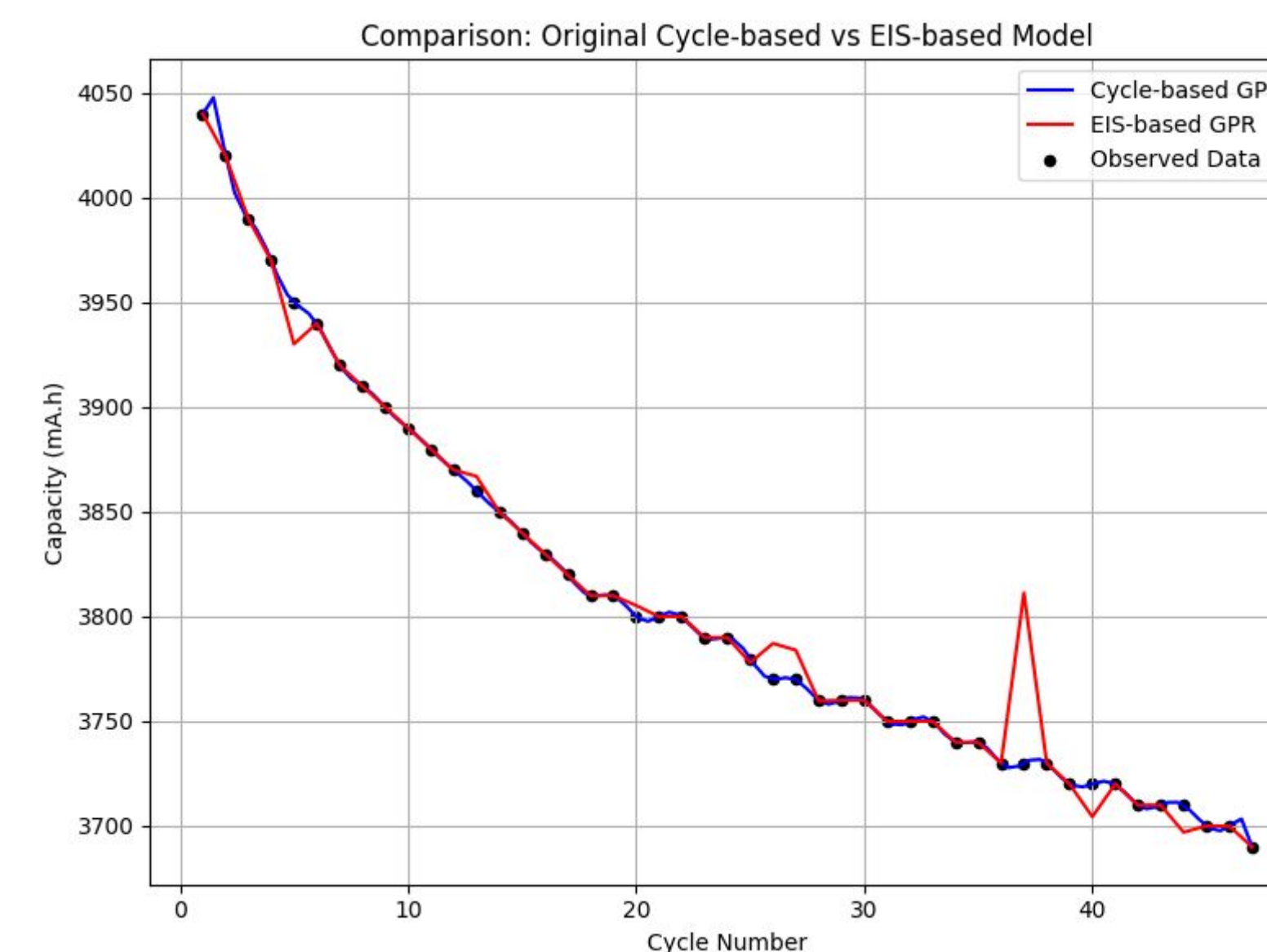


Fig 3: Capacity Degradation Over Cycles — EIS Model vs Ground Truth

Description: Plots model predictions vs actual capacity across cycles.

Explanation: EIS model closely follows real degradation, proving it can track battery aging.

Results & Discussion :

- The EIS-based GPR model achieved a high R^2 of 0.946 and low RMSE of 17.36 mAh, indicating strong prediction accuracy.
- Trained on real and imaginary impedance data from 69 frequencies.
- Outperformed a cycle-based baseline by capturing subtle battery aging patterns.
- Nyquist plots showed clear impedance evolution, enabling accurate capacity estimation.
- Mid-frequency signals (100–1000 Hz) contributed most to model performance.
- Offers a scalable, non-invasive solution for real-world battery health monitoring.